

LANGUAGE-SPECIALIZED MULTI-TEACHER ON-POLICY DISTILLATION FOR MULTILINGUAL LLM-BASED ASR

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ABSTRACT

Modern LLM-based ASR systems have established multilingual capability as a standard feature, leveraging large-scale multilingual corpora and LLMs’ cross-lingual knowledge to achieve competitive performance across multilingual benchmarks. However, joint modeling of languages with heterogeneous acoustic, phonological, and lexical characteristics inevitably introduces optimization conflicts, undermining language-wise specialization. To address this challenge, we propose **Language-Specialized Multi-Teacher On-Policy Distillation (LS-MOPD)**, which decouples language-specific knowledge acquisition from multilingual capability integration: language-specialized teachers are independently optimized via reinforcement learning (RL), after which their expertise is integrated into a generalist multilingual student through language routing and token-level multi-teacher distillation, thereby reducing direct cross-lingual optimization conflicts. We further explore two acoustic-prefix configurations, static and dynamic, to examine how teacher–student prefix consistency influences the efficacy of on-policy distillation. Experiments on benchmarks covering Mandarin, Mandarin subdialects, Cantonese, and English demonstrate that LS-MOPD substantially outperforms RL baselines and consistently surpasses the empirical performance envelope defined by best-performing RL teachers, revealing its potential to **generalize beyond all teachers** in multilingual ASR.

Index Terms— multilingual ASR, speech large language models, multi-teacher on-policy distillation

1. INTRODUCTION

LLM-based ASR has rapidly advanced as the mainstream paradigm in speech recognition, capitalizing on the extensive model capacity and rich linguistic priors of pretrained LLMs to achieve impressive performance across dozens of languages and dialects. Frontier systems such as Seed-ASR [1], Fun-ASR [2], and Qwen3-ASR [3] primarily rely on scaling up multilingual corpora, consolidating broad language capabilities into a generalist model in a data-driven manner.

Despite advances on multilingual ASR benchmarks, consolidating cross-lingual capabilities within a generalist model remains challenging. Languages and dialects differ systematically in their phonological, acoustic, and linguistic properties. For instance, Mandarin is characterized by pervasive homophony and thus relies heavily on contextual semantic disambiguation, whereas English exhibits considerable variability in pronunciation and prosody, demanding precise modeling of stress-timed rhythm and connected-speech phenomena such as phonetic reduction and cross-word linking. Jointly modeling such heterogeneous languages inevitably introduces gradient conflicts, leading to performance trade-offs across languages. This challenge is amplified in real-time ASR systems, where stringent latency constraints limit model scale, requiring heterogeneous multilingual capabilities to be accommodated within a finite model capacity budget.

Recent advances in multi-teacher on-policy distillation (MOPD) [4, 5] offer a principled remedy to this challenge. MOPD first trains a set of task-specialized teachers and distills their expertise into a student through task-routed supervision on the student’s on-policy rollouts [6, 7]. In multilingual ASR, task routing can be naturally adapted to language routing, decoupling language-specific optimization from multilingual integration. Heterogeneous language-wise objectives are independently optimized by specialized teachers and then distilled into the student, mitigating direct cross-lingual optimization conflicts. However, extending MOPD to LLM-based ASR raises several previously unexplored issues. Unlike text-only LLMs, LLM-based ASR conditions decoding on continuous acoustic prefixes produced by the encoder–adaptor. The teacher–student prefix consistency depends on the teacher training strategy and may directly affect distillation efficacy. Moreover, cross-lingual heterogeneity is generally less pronounced than cross-task heterogeneity, leaving it unclear whether aggregating supervision from multiple language-specialized teachers can yield genuinely complementary gains in multilingual ASR.

To address these issues, we compare alternative MOPD variants and analyze their behavior from the perspectives of gradient alignment and multi-teacher complementarity. Based on these insights, we propose **Language-Specialized Multi-Teacher On-Policy Distillation (LS-MOPD)**. Specif-

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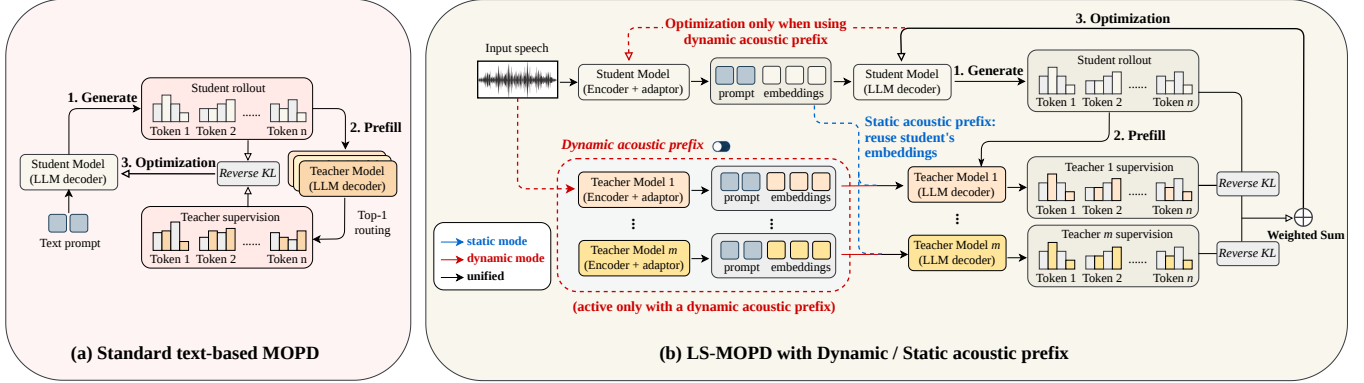


Fig. 1. Comparison of (a) standard text-based MOPD and (b) LS-MOPD for ASR under two acoustic-prefix configurations. In the dynamic configuration, the student and teachers use independently optimized encoder–adaptors and thus obtain model-specific acoustic prefixes. In the static configuration, all models share a frozen encoder–adaptor and therefore allow teachers to reuse the student’s acoustic prefixes. Both configurations support weighted multi-teacher supervision.

ically, we first train language-specialized teachers using the Decoupled Clip and Dynamic Sampling Policy Optimization (DAPO) algorithm [8] with an ASR-specific reward. During distillation, each student-generated on-policy rollout is routed to top- K teachers for the corresponding input-language category. The token-level supervision signals are then weighted and aggregated to distill complementary expertise into the generalist student. Our contributions are as follows:

- As the first study to explore MOPD for ASR, we present a systematic framework comprising DAPO-based teacher optimization, acoustic-prefix configurations, and multi-teacher weighted aggregation, offering practical guidance for effective distillation.
- We validate the effectiveness of LS-MOPD through analyses of gradient alignment and multi-teacher complementarity across distinct configurations.
- We empirically demonstrate the advantages of LS-MOPD over RL baselines and its potential to generalize beyond all teachers.

2. RELATED WORK

Multilingual ASR. Extending multilingual capability has become a major focus of modern ASR systems, with existing efforts following two main directions. The first leverages industrial-scale data expansion: systems such as Seed-ASR [1], Fun-ASR [2], Qwen3-ASR [3], FireRedASR2S [9], and NIM4-ASR [10] achieve strong multilingual performance by combining large-scale multilingual speech corpora with LLMs’ cross-lingual priors. However, phonological and lexical heterogeneity across languages generally demands greater model capacity, whereas real-time ASR systems must operate under stringent latency and computational constraints. This tension is reflected in deployment practice, where providers release multilingual-optimized variants,

such as Seed-ASR (ML) [1] and Fun-ASR-MLT-nano [2], alongside their flagship models. The second direction introduces language-specific architectural mechanisms, such as language-aware adaptors, mixture-of-experts routing, and gated query banks [11, 12, 13]. While these methods can improve multilingual performance at lower cost, they do not directly enhance the model’s underlying acoustic or semantic modeling capacity and may incur additional deployment overhead under high-concurrency inference.

Multi-teacher on-policy distillation. On-policy distillation (OPD) [14] has recently emerged as an effective LLM post-training algorithm, providing token-level supervision on the student’s sampled trajectories to mitigate exposure mismatch. This property makes OPD well suited to LLM-based ASR systems, which perform autoregressive generation yet require strict token-level transcription accuracy. Multi-teacher OPD [4, 5] extends this idea by independently optimizing multiple domain-specialized teachers and distilling them into the student model with reverse-KL-style objectives, making it suitable for capability integration. This is particularly appealing for multilingual ASR, as different languages rely on distinct acoustic-phonetic perception and lexical-semantic modeling capabilities. To the best of our knowledge, we are the first to introduce MOPD into the ASR domain.

3. METHODS

3.1. Language-Specialized RL Teacher Training

Prior to conducting MOPD, we construct a suite of teacher models specialized for distinct language categories. Starting from a base model π_{θ_0} built upon the encoder–adaptor–LLM architecture, we perform reinforcement learning with verifiable rewards (RLVR) based on DAPO [8]. For a given utterance $\mathbf{x}$ with reference transcription y , the rollout policy $\pi_{\theta_{\text{old}}}$ samples $G = 8$ hypotheses, $\{\tau_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | \mathbf{x})$. Fol-

lowing the reward design adopted in [10], each hypothesis is scored by an accuracy reward:

$$R_i = R_{\text{acc}}(\tau_i, y) = \exp[-2 \cdot \text{ER}_{\ell(\mathbf{x})}(\tau_i, y)], \quad (1)$$

where $\ell(\mathbf{x})$ denotes the language category of the input and $\text{ER}_{\ell(\mathbf{x})}$ denotes the language-dependent transcription error rate, instantiated as character error rate (CER) for Chinese and word error rate (WER) for English. The group-normalized advantage estimate is given by $\hat{A}_i = (R_i - \bar{R})/(\sigma_R + \varepsilon)$, where $\bar{R}$ and σ_R are the mean and standard deviation of the rewards within the rollout group, respectively. Groups with $\sigma_R = 0$ are discarded through dynamic sampling. DAPO then optimizes the following policy-gradient objective:

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E} \left[\frac{\sum_{i=1}^G \sum_{t=1}^{|\tau_i|} \min(r_{i,t} \hat{A}_i, \tilde{r}_{i,t} \hat{A}_i)}{\sum_{i=1}^G |\tau_i|} \right], \quad (2)$$

where $r_{i,t} = \pi_{\theta}(\tau_{i,t} | \mathbf{x}, \tau_{i,<t}) / \pi_{\theta_{\text{old}}}(\tau_{i,t} | \mathbf{x}, \tau_{i,<t})$ is the token-level importance ratio, and $\tilde{r}_{i,t} = \text{clip}(r_{i,t}, 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}})$ is its asymmetrically clipped counterpart. Here, ε_{low} and $\varepsilon_{\text{high}}$ denote the clipping margins. Compared with classic RLVR objectives, DAPO uses a token-level objective to balance contributions across transcription lengths and prevent errors in long utterances from being underweighted. Its dynamic sampling strategy further improves efficiency by resampling informative instances.

We construct a fixed multilingual pool of 50k utterances: 20k Mandarin, 10k Mandarin subdialects, 10k Cantonese, and 10k English samples. We group these data into three routing categories: Mandarin, Chinese dialects (Mandarin subdialects and Cantonese), and English. Each specialist teacher is trained on all data from its target routing category together with a randomly sampled 20% subset from each of the other routing categories, yielding three language-specialized teachers. In addition, a generalist teacher is trained on the full pool. For controlled comparison, DAPO training varies the acoustic front end along two orthogonal dimensions: encoder mode (streaming vs. offline) and encoder-adaptor trainability (frozen vs. trainable), resulting in four configurations: {streaming, offline} $\times$ {frozen, trainable}.

3.2. Language-Routed Multi-teacher OPD

After obtaining the three language-specialized teachers and the generalist teacher, we assess all teachers on held-out validation sets and rank them separately for Mandarin, Chinese dialects (including Mandarin subdialects and Cantonese), and English. Let $\pi_{\phi_{\ell(\mathbf{x}),k}}^T$ denote the k -th ranked teacher for routing category $\ell(\mathbf{x})$, where $k = 1$ denotes the best-performing teacher. The MOPD student π_{θ} is initialized from the base model π_{θ_0} . Following MOPD [5], the student generates an on-policy trajectory $\tau \sim \pi_{\theta}(\cdot | \mathbf{x})$. Let $s_t = (\mathbf{x}, \tau_{<t})$ denote

the decoding state at step t . The distillation loss from the k -th ranked teacher is defined as

$$\mathcal{L}_k(\mathbf{x}; \theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\frac{1}{|\tau|} \sum_{t=1}^{|\tau|} D_{\text{KL}} \left(\pi_{\theta}(\cdot | s_t) \parallel \pi_{\phi_{\ell(\mathbf{x}),k}}^T(\cdot | s_t) \right) \right], \quad (3)$$

and the final MOPD objective aggregates contributions from the top- K teachers ($K = 3$ by default):

$$\mathcal{L}_{\text{MOPD}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\sum_{k=1}^K \lambda_k \mathcal{L}_k(\mathbf{x}; \theta) \right], \quad \sum_{k=1}^K \lambda_k = 1. \quad (4)$$

The weights $\{\lambda_k\}$ govern the relative influence of each ranked teacher, while setting $\lambda_k = 0$ allows the effective number of active teachers to range from three to one. The formulation recovers top-1 teacher distillation (see Fig. 1 (a)) by setting $(\lambda_1, \lambda_2, \lambda_3) = (1, 0, 0)$, while also supporting weighted multi-teacher aggregation such as $(0.6, 0.2, 0.2)$. Importantly, the supervising teacher $\pi_{\phi_{\ell(\mathbf{x}),k}}^T$ is determined according to the evaluation performance of all candidate teachers for the language category $\ell(\mathbf{x})$. This ensures that each student trajectory can be supervised by the empirically strongest teachers for its target language category.

3.3. Preliminary Analysis on the Effect of Acoustic Prefix

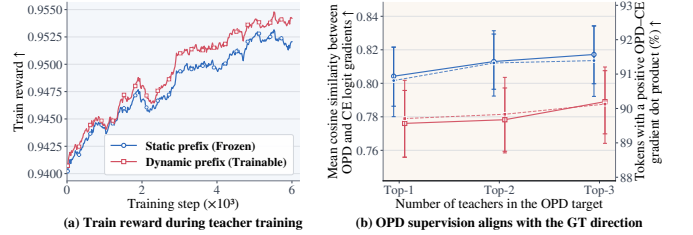


Fig. 2. Comparison of static and dynamic acoustic-prefix configurations in terms of (a) accuracy reward trajectories during RL and (b) OPD–CE gradient alignment metrics during distillation. Error bars are estimated from 10k bootstrap resamples.

When adapting MOPD to ASR, the LLM is conditioned on acoustic embeddings that serve as acoustic prefixes, whose configuration directly influences distillation effectiveness. A static acoustic-prefix configuration ensures prefix consistency between teachers and the student, mitigating the teacher–student conditioning mismatch during distillation. However, this configuration requires the teacher and student to share a frozen encoder–adaptor throughout RL and OPD, which limits the attainable performance of teachers. In contrast, a dynamic configuration allows unrestricted joint optimization of all model components, but at the risk of increased conditioning inconsistency.

To illustrate this trade-off, we conduct a preliminary analysis under static and dynamic acoustic-prefix configurations.

Table 1. Offline ASR performance measured by CER (%) for Chinese and WER (%) for English (lower is better). Model sizes are shown in brackets.

Method	WeNet		KeSpeech	WeNet-Yue		LibriSpeech		Avg.
	meet.	net	test	short	long	clean	other	
<i>Advanced Open-source Models</i>								
[0.8B] Fun-ASR-Nano [2]	4.68	5.22	7.18	7.31	10.02	1.63	4.35	5.77
[2.0B] Qwen3-ASR-1.7B [3]	4.00	4.13	4.98	5.79	8.00	1.56	3.49	4.56
[8B] Step-Audio2-mini [15]	4.23	4.63	3.98	7.78	8.44	1.22	2.61	4.70
[30B/A3B] Qwen3-Omni-Inst [16]	3.92	3.85	5.96	6.97	8.60	1.15	2.38	4.69
[8B] MiMo-V2.5-ASR [17]	4.63	4.52	7.76	6.53	10.52	1.35	3.42	5.53
<i>Our Methods</i>								
[2.3B] Base Model (NIM4-ASR)	5.01	4.80	4.57	5.22	9.48	1.23	2.61	4.70
<i>with static acoustic prefix</i>								
RL (Generalist Teacher)	4.88	4.74	4.40	5.27	9.49	1.18	2.58	4.65
RL (Best-Teacher Oracle)	4.77	4.70	4.40	5.19	9.30	1.18	2.56	4.59
LS-MOPD with 1 teacher	4.62	4.53	4.06	5.19	9.12	1.15	2.57	4.46
LS-MOPD with 2 teachers	4.60	4.52	4.13	5.17	9.16	1.15	2.53	4.47
LS-MOPD with 3 teachers	4.60	4.53	4.08	5.14	9.14	1.12	2.56	4.45
<i>with dynamic acoustic prefix</i>								
RL (Generalist Teacher)	4.56	4.69	4.49	5.35	9.55	1.20	2.57	4.63
RL (Best-Teacher Oracle)	4.56	4.69	4.45	5.25	9.19	1.17	2.56	4.55
LS-MOPD with 1 teacher	4.60	4.55	4.14	5.28	9.15	1.14	2.56	4.49
LS-MOPD with 2 teachers	4.49	4.58	4.13	5.23	9.16	1.17	2.48	4.46
LS-MOPD with 3 teachers	4.52	4.62	4.12	5.17	9.27	1.15	2.51	4.48

Note: Boldface separately indicates the best result among “Advanced Open-source Models” and among “Our Methods” for each benchmark.

As shown in Fig. 2, we compare (a) the accuracy-reward trajectories during RL training and (b) token-level alignment between the OPD gradients and the cross-entropy (CE) gradients derived from ground-truth labels during distillation. The latter is quantified by the mean cosine similarity and the proportion of directionally aligned tokens, defined as those with a positive OPD–CE gradient dot product. The results clearly show that the static prefix yields slightly lower teacher performance but better-aligned distillation gradients, whereas the dynamic prefix improves teacher performance at the expense of slightly reduced gradient alignment. This reveals a trade-off between teacher strength and distillation compatibility, further examined in the following section.

4. EXPERIMENTS

4.1. Experimental Setup

The base student model in this work is built upon an in-house backbone pretrained on approximately 560k hours of labeled speech, comprising a 0.6B FireRed Conformer [9], a $4\times$ downsampling linear adaptor, and a Qwen3-1.7B LLM decoder. For RL and MOPD, we construct a 50k-utterance multilingual training pool by sampling from representative datasets: 20k Mandarin utterances from WenetSpeech [18], 10k Mandarin subdialect utterances from KeSpeech [19], 10k Cantonese utterances from WenetSpeech-Yue [20], and 10k English utterances from LibriSpeech [21]. The generalist teacher and MOPD stages use the full multilingual pool, whereas language-specialized teachers are trained on their respective language-skewed subsets constructed with sampling ratios specified in Section 3.1.

For evaluation, we employ the official test sets of the

Table 2. Streaming ASR performance measured by CER (%) for Chinese and WER (%) for English (lower is better).

Method	WeNet		KeSpeech	WeNet-Yue		LibriSpeech		Avg.
	meet.	net	test	short	long	clean	other	
Base Model (NIM4-ASR)	6.19	5.26	5.69	6.06	12.33	1.34	3.32	5.74
<i>with static acoustic prefix</i>								
RL (Generalist Teacher)	5.92	5.19	5.54	5.68	11.10	1.29	3.27	5.43
RL (Best-Teacher Oracle)	5.84	5.18	5.54	5.50	11.10	1.25	3.20	5.37
LS-MOPD with 1 teacher	5.76	5.00	5.12	5.61	10.83	1.25	3.21	5.25
LS-MOPD with 2 teachers	5.67	4.99	5.17	5.59	10.67	1.24	3.15	5.21
LS-MOPD with 3 teachers	5.69	4.96	5.09	5.52	10.61	1.23	3.19	5.18
<i>with dynamic acoustic prefix</i>								
RL (Generalist Teacher)	5.90	5.26	5.52	5.60	11.10	1.35	3.22	5.42
RL (Best-Teacher Oracle)	5.72	5.26	5.50	5.58	10.96	1.31	3.16	5.36
LS-MOPD with 1 teacher	5.65	5.17	5.30	5.58	11.19	1.29	3.17	5.34
LS-MOPD with 2 teachers	5.71	5.16	5.24	5.99	11.02	1.27	3.12	5.36
LS-MOPD with 3 teachers	5.78	5.26	5.22	5.51	10.90	1.28	3.15	5.30

aforementioned four datasets, comprising seven subsets in total. Across all evaluations, a unified text normalization pipeline based on WenetTextProcess is applied to mitigate the effects of surface-form variations. All experiments adopt deterministic decoding strategies. In the streaming setting, speech is encoded in 640 ms chunks, with four preceding chunks retained as left context, and greedy decoding (beam size = 1) is employed. In the offline setting, the encoder operates over the complete utterance, and beam search decoding is applied with a beam size of 3.

4.2. Training Details

For RL training, we train for at most 6k optimization steps with a maximum learning rate of 2×10^{-6} linearly decayed to zero, and a sampling temperature that is cosine-annealed from 1.0 to 0.7. For MOPD, we train for 20k steps using a maximum learning rate of 4×10^{-6} cosine-annealed to 2×10^{-6} and a fixed sampling temperature of 1.0. All RL and MOPD experiments are conducted on 8 NVIDIA A100 GPUs with BF16 precision and DeepSpeed ZeRO-2 [22].

4.3. Main Results

Tables 1 and 2 report the performance of our proposed methods on benchmarks, alongside leading open-source models as references for the current state of the art. Within each acoustic-prefix configuration, the generalist teacher serves as the RL baseline, whereas the best result attained by any candidate teacher on each benchmark is reported as “RL (Best-Teacher Oracle)”. For LS-MOPD, the number of supervising teachers is determined by the loss weights in Eq. (4): the one-, two-, and three-teacher variants use $\mathcal{L}_{\text{MOPD}} = \mathcal{L}_1$, $\mathcal{L}_{\text{MOPD}} = 0.6\mathcal{L}_1 + 0.4\mathcal{L}_2$, and $\mathcal{L}_{\text{MOPD}} = 0.6\mathcal{L}_1 + 0.2\mathcal{L}_2 + 0.2\mathcal{L}_3$, respectively. We analyze the results as follows:

(1) Overall Performance. Using only a 50k-utterance post-training pool, LS-MOPD substantially improves upon the base model and consistently outperforms the RL baseline. The average error rate drops to a minimum of 4.45%, outperforming all open-source baselines, including models trained

on million-hour-scale speech corpora [2, 3] and those built upon larger LLM backbones [15, 16, 17], using only 2.3B parameters. Notably, LS-MOPD surpasses the empirical performance envelope defined by the best-teacher oracle on almost all benchmarks, and achieves the best results among all compared leading systems on WeNetSpeech-Yue-short and LibriSpeech test-clean. These results demonstrate that the distilled student can outperform all teachers, achieving highly competitive multilingual recognition accuracy together with measurable emergent gains. As shown by the streaming results in Table 2, LS-MOPD also delivers substantial gains in the streaming setting, with relative improvements exceeding those observed in the offline configuration.

(2) Static vs. dynamic acoustic prefix. Before distillation, dynamic-prefix RL teachers achieve slightly better average performance than their static-prefix counterparts, owing to joint encoder–adaptor optimization that yields more refined acoustic representations. After MOPD, however, static-prefix variants attain the best average results in both the offline and streaming settings. We attribute this reversal to teacher–student acoustic-prefix mismatch under the dynamic configuration: when teacher supervision depends on acoustic cues absent from the student’s representation, distillation may shift from learnable preference transfer toward fitting partially unreachable targets.

(3) Top- K teacher weighted aggregation. Since cross-lingual heterogeneity is generally less pronounced than cross-task heterogeneity, we extend the default top-1 routing of MOPD to weighted top- K teacher aggregation. This yields modest gains, particularly in the streaming setting, suggesting a beneficial regularization effect that becomes more pronounced under higher decoding uncertainty. Notably, we find that incorporating the second- and third-ranked teachers can lead to improvements despite their substantially lower standalone performance than the best teacher, suggesting that teacher diversity can provide complementary supervision and alleviate token-level overconfidence.

4.4. Sources of Improvement and Emergent Gains

Beyond benchmark results, we conduct visual analyses to validate LS-MOPD and offer insight into its emergent behavior.

As shown in Fig. 3 (a), we examine the relationship between the benchmark gains of different LS-MOPD variants and the proportion of tokens with aligned OPD and CE gradients, as defined in Section 3.3. The results show a positive association: static-prefix LS-MOPD variants exhibit more consistently aligned gradients and achieve larger performance gains. By contrast, although dynamic-prefix teachers show stronger standalone performance, their supervision induces gradients that are less aligned with the CE gradients, leading to smaller distillation gains. Moreover, multi-teacher supervision generally yields better gradient alignment than single-teacher supervision. This finding provides a preliminary

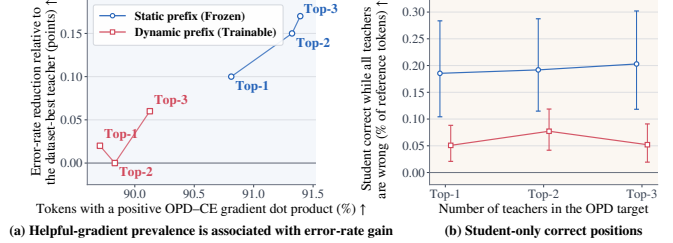


Fig. 3. (a) Relationship between OPD–CE gradient alignment and the performance gains achieved by LS-MOPD. (b) Emergent gains: the proportion of tokens incorrectly predicted by all teachers but correctly predicted by the student.

mechanistic explanation for why multi-teacher distillation can outperform its single-teacher counterpart: lower-ranked teachers can still contribute complementary supervision that induces more favorable optimization directions.

Fig. 3 (b) further reveals the emergent benefit of LS-MOPD: a non-negligible proportion of tokens are predicted incorrectly by all teachers, yet are correctly recognized by the student. Together with the consistent improvements of LS-MOPD over the best-teacher oracle in Tables 1 and 2, these results highlight the potential of LS-MOPD for multilingual ASR, demonstrating that coordinated supervision from multiple teachers can enable the student to generalize beyond the performance envelope of any individual teacher.

5. CONCLUSION

This work proposes a multi-teacher on-policy distillation framework for multilingual LLM-based ASR. We systematically investigate the impact of static and dynamic acoustic-prefix configurations and integrate language-specialized teachers through language-routed, weighted supervision. Experimental results across both offline and streaming scenarios demonstrate that LS-MOPD substantially outperforms RL baselines and has the potential to generalize beyond the empirical upper envelope of trained RL teachers. We further analyze the effectiveness of LS-MOPD from the perspectives of gradient alignment and multi-teacher complementarity, and derive practical insights into acoustic-prefix selection and multi-teacher aggregation.

Our work also has several limitations. First, the evaluation is relatively narrow in scope, covering only a single model backbone and a limited set of target languages. Broader validation across model architectures and languages is still warranted. Second, our current dynamic acoustic-prefix configuration remains preliminary and does not yet fully exploit the potential of this design. Future work will explore teacher–student supervision mechanisms that preserve the optimization flexibility of dynamic acoustic prefixes while reducing the resulting conditioning mismatch.

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